

CS5070 Mathematical Structures for Computer Science

Logic & Set

Kun Suo

Department of Computer Science
College of Computing and Software Engineering
Kennesaw State University

ksuo@kennesaw.edu

Basic Definitions

Statement

- A declarative sentence that is true or false. It is also known as a proposition.

Sentence	Statement?	Why
Atlanta is in Georgia.	Yes	True
$2 + 2 = 5$	Yes	False
Close the door.	No	Command
What time is it?	No	Question
$x > 5$	No	Truth depends on x

Atomic Statement

- A statement that cannot be divided into smaller statements.

Examples:

- P: "The server is online."
- Q: "The database is reachable."

Molecular (or Composite) Statement

- A statement that is not atomic.

Examples:

- "The server is online AND the database is reachable."
- "The server is online OR the database is reachable."

Basic Definitions..

Logical Connectives

- Are used to define more complicated statements (molecular) out of simpler ones atomic or molecular).

Binary Connectives

- Used to connect two statements. These are: **and, or, xor**.

Unary Connectives

- Applied to a single statement. This is **not**; also known as negation.

Propositional Variables

- Represent statements. These are denoted by P, Q, R, S, ..

Logical Connectives

The logical connectives are: $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$, $\neg$. Examples:

$P \wedge Q$, known as conjunction

$P \vee Q$, known as disjunction

$P \oplus Q$, known as exclusive or

$P \Rightarrow Q$, known as implication (or conditional)

$P \Leftrightarrow Q$, known as biconditional (if and only if)

$\neg P$, known as negation

The truth value of a statement is determined by the truth values of its parts

Truth Conditions for Connectives

$P \wedge Q$ is true when both P and Q are true (logical and)

$P \vee Q$ is true when P or Q , or both are true, (inclusive or)

$P \oplus Q$ is true when exactly one P or Q are true. and is false otherwise (exclusive or)

$P \Rightarrow Q$ is true when P is false or Q is true or both are true

$P \Leftrightarrow Q$ is true when P and Q are both true, or both false

$\neg P$ is true when P is false

Example

Suppose:

P = "I submit the assignment."

Q = "I pass the course."

Conjunction (AND)

$P \wedge Q$

"I submit the assignment and I pass the course."

Only true when BOTH are true.

Disjunction (OR)

$P \vee Q$

"I submit the assignment or I pass the course."

Inclusive OR.

If I do both $\rightarrow$ true.

XOR

$X \oplus Y$

"I choose exactly one tea or coffee."

If I drink both $\rightarrow$ false.

Negation

$\neg P$

"I did not submit the assignment."

Example

Suppose:

P: "It is raining."

Q: "The ground is wet."

Implication: $P \Rightarrow Q$

If it is raining, then the ground is wet.

P (Raining)	Q (Ground Wet)	($P \rightarrow Q$)
T	T	T
T	F	F
F	T	T
F	F	T

Case 1: Raining and ground is wet

Promise kept.

True.

Case 2: Raining but ground is NOT wet

This violates the statement.

False.

Case 3: Not raining but ground is wet

Maybe a sprinkler was used.

The statement never said the ground can only be wet when it rains.

True.

Case 4: Not raining and ground not wet

Nothing contradicts the statement.

True.

Example

Suppose:

P: "You know the correct password."

Q: "You can log in."

Biconditional: $P \Leftrightarrow Q$

You can log in if and only if you know the correct password.

P	Q	$(P \leftrightarrow Q)$
T	T	T
T	F	F
F	T	F
F	F	T

Case 1: Know password and can log in

Matches perfectly.

True.

Case 2: Know password but cannot log in

Something is wrong.

False.

Case 3: Don't know password but can log in

Security problem.

False.

Case 4: Don't know password and cannot log in

Matches perfectly.

True.

Implications

An implication or conditional is a molecular (compound) statement of the form:

$$P \implies Q$$

where P and Q are statements.

P is the hypothesis (or antecedent)

Q is the conclusion (or consequent)

An implication is true if P is false or Q is true (or both), and false otherwise

The only way for $P \implies Q$ to be false is for P to be true and Q to be false

Direct Proofs of Implications

To prove an implication $P \Rightarrow Q$, it is enough to assume P , and from it, deduce Q .

Explain why Q is true, but you get to assume P is true first

Prove: If two numbers a and b are even, then their sum is even

Proof. Suppose the numbers a and b are even. This means that $a = 2k$ and $b = 2j$ for some integers k and j . The sum is then $a + b = 2k + 2j = 2(k + j)$. Since $k + j$ is an integer, this means that $a + b$ is even. ■

DeMorgan's Law

Rules in logic that explain how **NOT** interacts with **AND** and **OR**.

It is NOT true that I am
both eating pizza and
drinking soda.

P: "I am eating pizza."
Q: "I am drinking soda."

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

This could happen if:

I'm not eating pizza, OR
I'm not drinking soda, OR
neither.

Today is neither
Saturday nor Sunday.

P: "Today is Saturday."
Q: "Today is Sunday."

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

This could happen if:

Today is not Saturday,
AND
Today is not Sunday.

Converse and Contrapositive

The **converse** of an implication $P \Rightarrow Q$, is the implication $Q \Rightarrow P$

The converse is not logically equivalent to the original implication.
That is, whether the converse of an implication is true is independent of the truth of the implication

The **contrapositive** of an implication $P \Rightarrow Q$ is the statement

$$\neg Q \Rightarrow \neg P$$

An implication and its contrapositive are logically equivalent (they are both true or both false)

Understanding converses and contrapositives can help understand implications and their truth values

The contrapositive always has the same truth value as its original implication.

Biconditional

When the implication $P \Rightarrow Q$ and the implication $Q \Rightarrow P$ are both true, then P and Q are equivalent and is denoted as $P \Leftrightarrow Q$

The biconditional statement $P \Leftrightarrow Q$ is logically equivalent to $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$

You can think of a biconditional (if and only if) statement as having two parts: an implication and a converse

" P is necessary for Q " means $Q \Rightarrow P$

" P is sufficient for Q " means $P \Rightarrow Q$

" P is necessary and sufficient for Q ", then $P \Leftrightarrow Q$

Example

Example:

P = "Number is divisible by 2"

Q = "Number is even"

$P \leftrightarrow Q$

True both directions.

Counterexample:

P = "Person is a professor"

Q = "Person has a PhD"

Not equivalent.

Many PhD holders are not professors.

Necessary vs Sufficient

Sufficient

Being a dog is sufficient for being an animal.

$\text{Dog} \rightarrow \text{Animal}$

Necessary

Being an animal is necessary for being a dog.

$\text{Dog} \rightarrow \text{Animal}$

Necessary and Sufficient

Even number $\leftrightarrow$ divisible by 2

Truth Tables

These tables show the truth value of outcome for every combination of the various truth values of the sentences (propositions)

P	Q	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
t	t	t	t	t	t
t	f	f	t	f	f
f	t	f	t	t	f
f	f	f	f	t	t

Predicates and Quantifiers

A sentence that contains variables is known as a predicate. For example, $P(x)$ and $Q(y)$

The **existential quantifier** is denoted as $\exists$ and means 'there exists'. For example,

$$\exists x (x < 0)$$

asserts that there is a number less than 0

The **universal quantifier** is denoted as $\forall$ and means 'for all'. For example,

$$\forall x (x \geq 0)$$

asserts that every number is greater than or equal to 0

Quantifiers and Negation

$\neg \forall x P(x)$ is equivalent to $\exists x \neg P(x)$

$\neg \exists x P(x)$ is equivalent to $\forall x \neg P(x)$

Example, prove:

$$\exists x \forall y (y \geq x)$$

We could quantify over natural numbers $(0, 1, 2, \dots)$

Another use of the universal quantifier:

$$\forall x (S(x) \implies R(x))$$

Sets

A set is an unordered collection of objects

Two sets are equal exactly if they contain the exact same elements

Notation: set **A** contains the elements 1, 2, 3

$$A = \{1, 2, 3\}$$

Notation: element **a** is a member of the set containing the elements a, b, c

$$a \in \{a, b, c\}$$

Notation: element **d** is not in the set containing elements a, b, c

$$d \notin \{a, b, c\}$$

Similarly, element 4 is not in set **A**

$$4 \notin A$$

Set of Natural Numbers

$\mathbb{N}$ is a set of natural numbers

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}$$

Let A be the set of all even natural numbers

$$A = \{0, 2, 4, 6, \dots\}$$

A better way to denote this is:

$$A = \{x \in \mathbb{N} : \exists n \in \mathbb{N} (x = 2n)\}$$

Another way

$$A = \{x \in \mathbb{N} \mid x \text{ is even}\}$$

The set of integers; positive and negative whole numbers is denoted as $\mathbb{Z}$. Using this, we can define the set:

$$A = \{x \in \mathbb{Z} : x^2 \in \mathbb{N}\}$$

Additional Notation

$\mathbb{N}$ is the set of natural numbers

$\mathbb{Z}$ is the set of integer numbers

$\mathbb{R}$ is a set of real numbers

$\mathbb{Q}$ is the of rational numbers

The empty set is denoted as $\emptyset$

The universe set, the set of all elements, is denoted as U

$P(A)$ is the power set of any set A

The cardinality of set A is denoted as $|A|$ and is the number of elements in A

Set Relationships

Given sets $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$, clearly:

$$A \neq B \text{ and } A \subset B$$

In simple words, sets A and B are not equal. Set A is a subset of set B because every element of A is also an element of B .

The cardinality of set A is 3 and the cardinality of set B is 4.

$$|A| = 3, \quad |B| = 4$$

The collection of all subsets of B is a new set, which is a set of sets. This set is known as the power set of B and denoted as $P(B)$

Given set $A = \{1, 2, 3\}$, the set of all subsets of A is:

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Set Operations

The union of two sets is the operation that combines two sets to get the collection of objects that are in either set.

$$C = A \cup B$$

C is a new set, the union of sets A and B.

The intersection of two sets A and B results in another set with the elements that are exactly the elements in both A and B

$$C = A \cap B$$

The complement of a set A is a set that has the elements not in set A. When set D is the complement of set A, it is denoted as:

$$D = \bar{A}$$

More Set Operations

The intersection of set **A** and complement of set **B** is denoted:

$$A \cap \bar{B}$$

Another way to write the previous operation is defined by the set difference as:

$$A \cap \bar{B} = A \setminus B$$

The set Cartesian product of sets **A** and **B** is defined by:

$$A \times B = \{(a, b) : a \in A \wedge b \in B\}$$

This is the set of all ordered pairs (a, b) such that elements a are in set **A** and elements b are in set **B**.

Cartesian Product

Given sets **A** and **B**, the following set is defined:

$$A \times B = \{(x, y) : x \in A \wedge y \in B\}$$

This is the set of all ordered pairs (x, y) where x is an element of **A** and y is an element of **B**. This is the Cartesian product of **A** and **B**.

Let $A = \{a_1, a_2, a_3, \dots, a_m\}$ and $B = \{b_1, b_2, b_3, \dots, b_n\}$

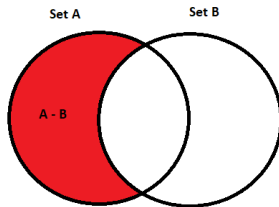
$$A \times B = \{$$

$$\begin{array}{ccccccc} (a_1, b_1), & (a_1, b_2), & (a_1, b_3), & \dots & (a_1, b_n), \\ (a_2, b_1), & (a_2, b_2), & (a_2, b_3), & \dots & (a_2, b_n), \\ (a_3, b_1), & (a_3, b_2), & (a_3, b_3), & \dots & (a_3, b_n), \\ \vdots & & & & \\ (a_m, b_1), & (a_m, b_2), & (a_m, b_3), & \dots & (a_m, b_n) \end{array}$$

$$\}$$

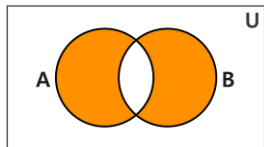
More Set Operations

The set difference $A \setminus B$ of sets A and B is the set that consists of all those elements in A that are not in set B .



More Set Operations

The symmetric difference of two sets **A** and **B**, also known as the disjunctive union, is the set of elements which are in either of the sets, but not in their intersection. This operation is denoted as



$$A \oplus B = A \triangle B = (A \setminus B) \cup (B \setminus A)$$

The main property is that an element in $A \oplus B$:

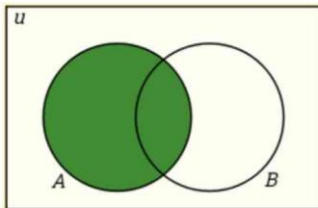
$$x \in A \oplus B \iff (x \in A \cup B \wedge x \notin A \cap B)$$

For example, the symmetric difference of the sets $A = \{1, 2, 3\}$ and $B = \{3, 4\}$ is

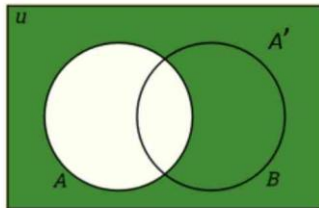
$$A \oplus B = \{1, 2, 4\}$$

Venn Diagrams

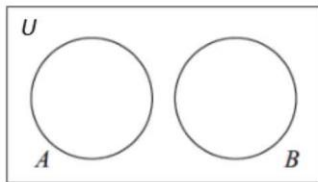
Set Operations and Venn Diagrams



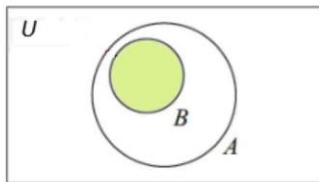
Set A



A' the complement of A

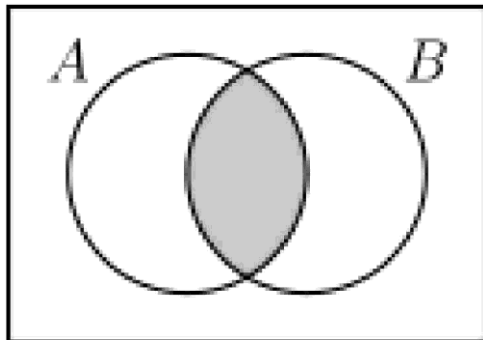


A and B are disjoint sets



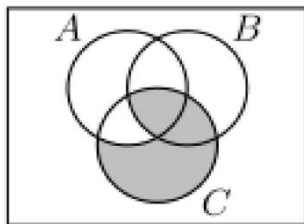
B is proper subset of A
 $B \subset A$

Venn Diagrams: Intersection



Venn Diagrams: More

$$(B \cap C) \cup (C \cap \bar{A})$$



Questions?